

King Mongkut's University of Technology Thonburi
Department of Computer Engineering

Course Syllabus

Class ID/ Credit	CPE 333 (TH) ระบบปฏิบัติการ (EN) Operating Systems 3 (3-0-6)
Lecturer	Asst.Prof. Rajchawit Sarochawikasit Email rajchawit.sar@mail.kmutt.ac.th
Teaching Assistant	Omar Yusoh
Building	LIB108
Schedule	Tue 13:30 – 16:20 (Lecture Based)

Description

Theoretical aspects of Operating systems: memory management, process management, I/O management, and information management. Issues on Job Control Language, Assembler, Loader, and Link. Example cases of operating systems, compilers, interpreters and utilities.

Understand the concept of operating systems, able to understand small, basic Linux kernel code.

Grade Distribution (Non-OBEM maps OBEM)

Midterm Exam	30% (Module A)
Final Exam	30% (Module B and C)
Mini-project	20% (Selected module A, B, and C)
Lab/Assignment/Quiz	20% (Module A, B, and C)

Grading criteria

Relative to the class peer performance. Usually a total score below 35% results in an F grade.

Textbook

1. **Operating Systems: Three Easy Pieces** by Arpaci and Dusseau
2. **Operating System Concepts (8th ed.)** by Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne. 2008. Wiley Publishing.

Schedule

Week #	Topic	Problem Session
1	Introduction to Operating Systems	No
2	Process and Process APIs	Yes
3	Scheduling	Yes
4	Contiguous Memory Allocation	Yes
5	Segmentation and Paging Schemes (Non-contiguous Memory Allocation)	No
6	Virtual Memory	Yes
7	<Mini-Project: Proposal Presentation>	N/A
	--- Mid Term Exam ---	N/A
8	Concurrency and Threads	Yes
9	Lock Mechanism	Yes
10	Condition Variables and Semaphores	No
11	Resource Allocation and Deadlock	Yes
12	I/O Devices and Mass Data Storage	No
13	File System and Implementation	Yes
14	<Mini-Project: Final Defend>	N/A
	--- Final Exam ---	N/A

Note: Module A in red, module B in blue, and module C in green.